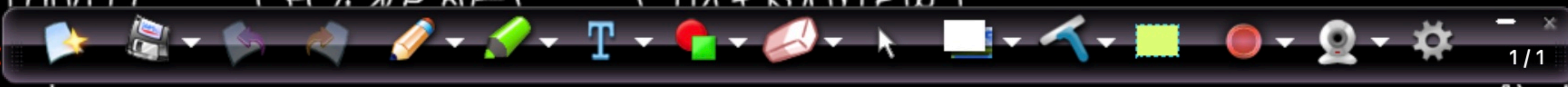
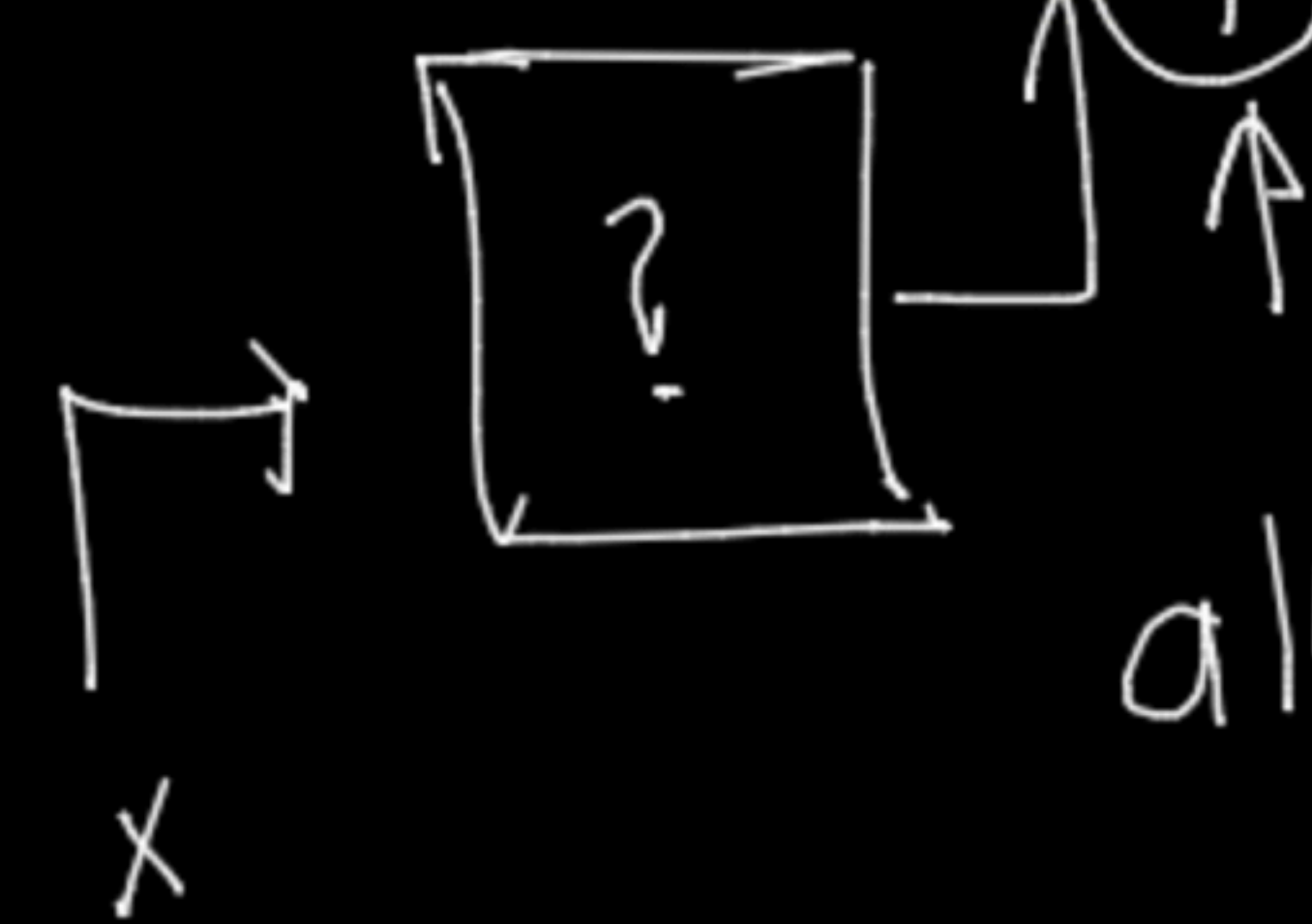


Bias - Variance Tradeoff (Interview)



→ theory in ML



(1000) input → qov

$k=1$

overfitting

$k=n$

underfitting

0.1%

many reasons?
10%

Assumption

Logistic Regression?

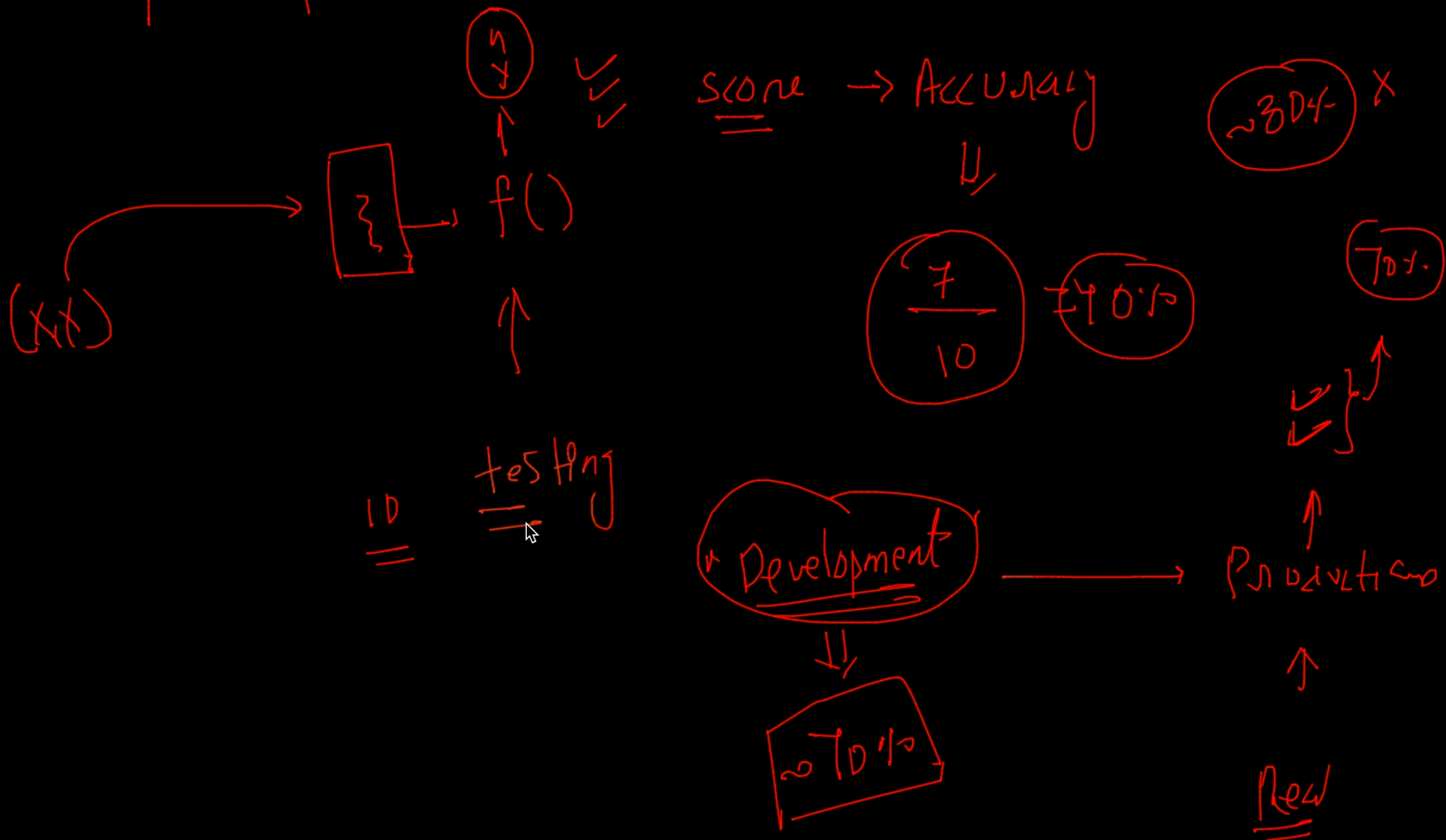
Data

using math → Bias-variance tradeoff

generalization error = Bias + variance + irreducible error

← (High-bias) → underfitting

(**) Performance measurement of model



$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

	(Aktuell)		(Prob-score)	(Prob-score)
X	Y	M1	M2	
X1	1 ✓	0.9	0.6	
X2	1 ✓	0.8	0.65	
X3	0 ✓	0.1	0.15	
X4	0 ✓	0.15	0.48	

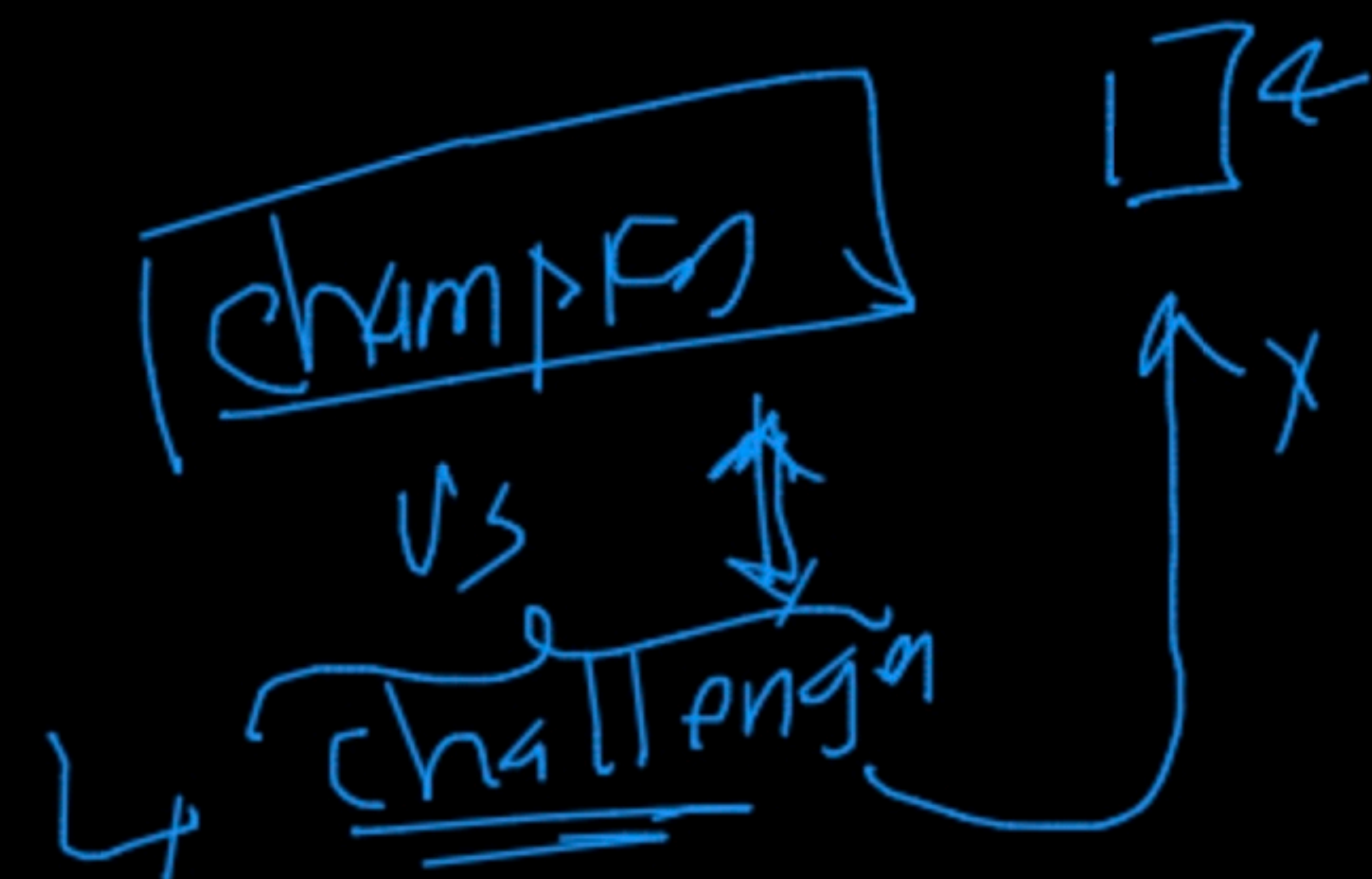
$\hat{Y}_1$	$\hat{Y}_2$		don't use <u>Act</u>
1 ✓	1 ✓		if <u>prob-score</u>
1 ✓	1 ✓		
0 ✓	0 ✓	M2 =	4/4 $\Rightarrow$ 100%
0 ✓	0 ✓		

$\textcircled{M1} \rightarrow \text{Acc} \Rightarrow \frac{4}{4} \Rightarrow \underline{100\%}$

if prob-score > 0.5

$\rightarrow 1$
else $\rightarrow 0$

prediction



$\begin{pmatrix} 0 & 1 \end{pmatrix}$
 $\downarrow$
0.10 0.90

0.57 $\rightarrow$ ①

II Confusion Matrix

↓ ↓ does not / can not process prob-score

0 → -ve

1 → +ve

Binary Classification
(0,1) ← Actual

testing data

	0	1
0	↑ 1 ?	↓ 1 ?
1	↓ 2	3 ↑

total testing data

→ 3 + 4 ⇒ 7

How many points which predicted value as '0' and actual value is also zero

Y	↑
1	1
1	0
1	1
0	0
0	0
0	1

ideal confusion matrix of a good model

prediction

	Actual	
	0	1
0	2 ↑	0 ↓
1	0 ↓	2 ↑

Sklearn's

v.v. good model

Y	Y
1	1
0	0
0	0

Extend → to n-class classification

	0	1	2	3
0	↑	↓	↓	↓
1	↓	↑	↓	↓
2	↓	↓	↑	↓
3	↓	↓	↓	↑

Actual

0 1

prediction

0	TN	FN
1	FP	TP

$N + P = \text{total data point}$

(i) TPR
(True + rate)

$\Downarrow$

$$= \frac{\#TP}{P}$$

$$(ii) TNR = \frac{\#TN}{N}$$

$$(iii) FPR = \frac{FP}{P}$$

$$(iv) FNR = \frac{FN}{N}$$

TP $\rightarrow$ what is predicted class

are you correct?

Examples

		Actual	
		0	1
Prediction	0	850	6
	1	50	94
		900	100 $\Rightarrow$ 1000

$$(iii) FPR = \frac{50}{900} \quad \downarrow \downarrow \quad \underline{\underline{\text{small}}}$$

$$(iv) FNR = \frac{6}{100} \Rightarrow 6\%$$

For any good model

$$(i) TPR = \frac{94}{100} \Rightarrow 94\%$$

$$(ii) TNR = \frac{850}{900} \Rightarrow \uparrow$$

$\left\{ \begin{array}{l} (TPR \text{ and } TNR) \rightarrow \uparrow \uparrow \uparrow \\ \text{and} \\ (FPR \text{ and } FNR) \rightarrow \downarrow \downarrow \end{array} \right.$

⇒ Precision, Recall, F1-score
↑

sklearn
↑

→ $P_r = \frac{TP}{TP + FP}$
↑

of all the points the model
predicted / declared to be
true, what % age of them
are actually true

(ii) Recall ⇒ $\underline{TPR} = \frac{TP}{P}$

(iii)

$$F1\text{-score} = 2 \times \left(\frac{P_n \times \text{Recall}}{P_n + \text{Recall}} \right)$$

Any good model

↑

$P_n \uparrow$ (i) $\text{Recall} \uparrow$ (ii)

F1 ↑